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**A Low-Noise High-Gain Broadband Compact Inverter-Based
Transimpedance Amplifier in 22-nm FDSOI CMOS.**

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Outline



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- I. Introduction
- II. Design Method
 - A. TIA with Active Feedback
 - B. TIA with Inductive peaking
- III. Layout Implementation and Fabrication
- IV. Measurements

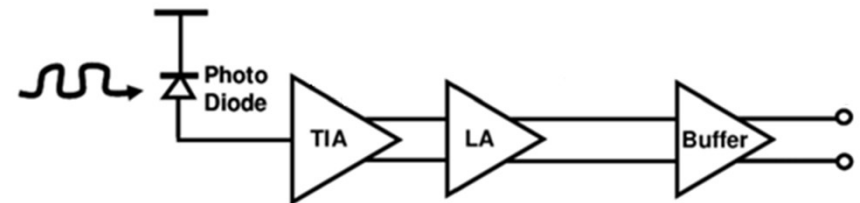
Introduction



The desired data rate has been exponentially increased up to tens of Gb/s.

Optical fibers : best medium for carrying large volumes of data, superior performance in terms of bandwidth, crosstalk, ...

- Transimpedance Amplifier (TIA):
 - First stage following the Photo-diode(PD)
 - High Gain, High Bandwidth, Low Noise
- Design challenge: capacitance of photo-diode and transistors.
- Nanometer CMOS : ultra-wide bandwidth due to their high unity-gain-frequency, f_T , low power consumption and cascading multiple gain stages Amplifier.



Introduction

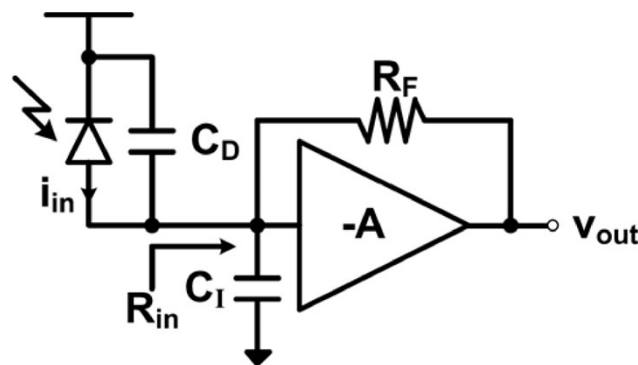


- **Common-gate (CG) TIA**

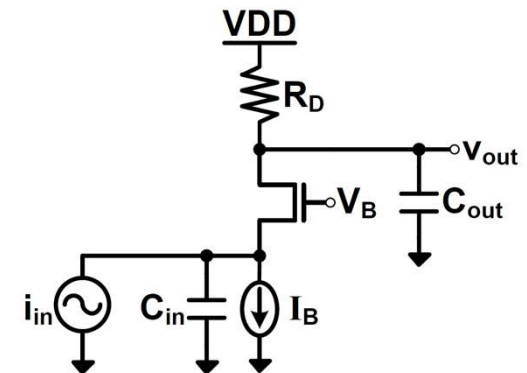
inherently have low input impedance and wider bandwidth, but they have higher noise.

- **shunt-feedback (SF) TIA**

provide higher gain and lower noise; however, their input impedance cannot be lowered enough to extend their bandwidth.



Feedback TIA



Common-Gate TIA

Introduction



- **shunt-feedback (SF) TIA**

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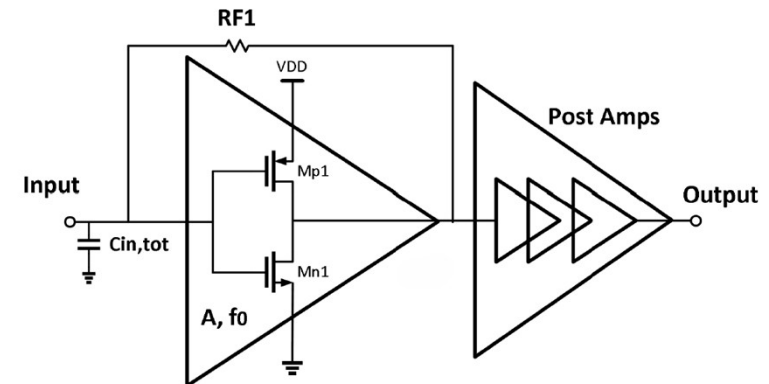
- **Inverter Based using CMOS**

- higher trans-conductance (g_m) because of NMOS and PMOS transistors together.
- Large gain cause higher $R_{F,T}$, which reduces the noise contribution.
- low voltage supplies.
- benefits from a cascade stage to enhances its bandwidth.

Introduction



- **SF TIA**
- The transimpedance gain of the SF TIA is inversely proportional to the square of the bandwidth.
- $A.f_0$ is the amplifier gain-bandwidth product and proportional to f_T .
- A and f_0 are the open loop gain and BW.
- Input referred noise:



$$R_T \leq \frac{A.f_0}{2\pi.C_{tot,in}.BW^2}.$$

$$R_T = \frac{A}{A+1}.R_{F1} \approx R_{F1}.$$

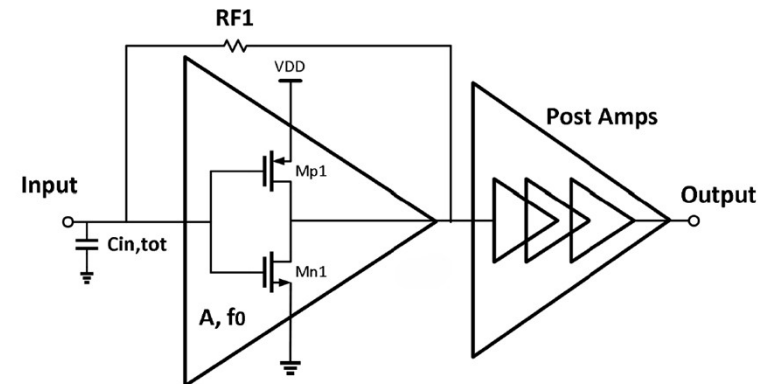
$$\overline{I_{n,in,SF}^2}(s) = \overline{I_{n,in,RF1}^2}(s) + \overline{I_{n,in,amp}^2}(s) + \overline{I_{n,in,post}^2}(s) = \frac{4kT}{R_{F1}} + \frac{4kT\gamma}{R_{F1}^2}|1 + sR_{F1}.C_{tot,in}|^2 + \frac{4KT\gamma/g_{m,post}}{|Z_{T,tot}(s)|^2}$$

Introduction



- **SF TIA**

- The first-stage TIA bandwidth is scaled down by a factor of n , the feedback resistance $R_{F,T}$ scales up n^2 times resulting in n^2 times reduction in feedback resistor current noise, n times increase in amplifier gain and n^4 times reduction in amplifier's white noise.
- Narrow-Band TIA is an approach to make TIA noise performance independent of the bandwidth : low-noise TIA followed by post amplifier chain in order to restore the required bandwidth [6].



$$R_T \leq \frac{A \cdot f_0}{2\pi \cdot C_{tot,in} \cdot BW^2}.$$

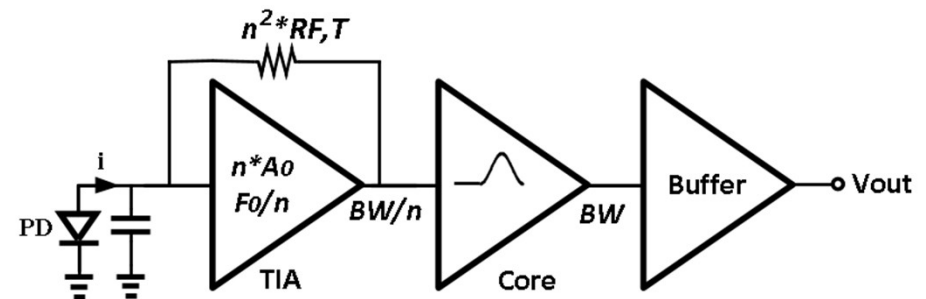
$$R_T = \frac{A}{A+1} \cdot R_{F1} \approx R_{F1}.$$

Design Method- TIA with AFB



inductor-less TIA

- Narrow-band inverter-based TIA for noise reduction.
- Core: Inverter-based gain block with active feedback technique (TIA-AFB) for bandwidth enhancement.
- Buffer stage: match to 50 ohm.
- realization in low supply technologies.
- Reduced Area.



Introduction



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- Bandwidth Extension Technique
 - Inductive peaking : shunt peaking, series peaking, T-coil.
 - Inductor-less Method: Active feedback, Active Inductor, multi peaking, positive feedback.



Design Method

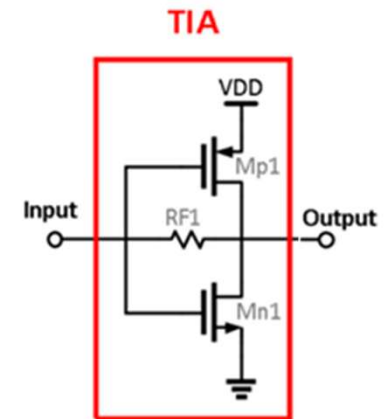
- Compared to existing inductor-less TIA solutions, the number of feedback loops and complexity of the core circuit in terms of stability and group delay flatness are reduced by utilizing two inverter stages with different sizing along with one inverter-based feedback loop.
- Different sizing of inverters in the core circuit results in two different pairs of complex conjugate poles, lowering the number of feedback loops while providing enough degree of freedom to optimize the TIA response.
- This approach results in a low-power design with minimal area, low noise, and enhanced bandwidth.

Design Method- TIA with AFB



Target for Narrow-band TIA

- BW for the entire TIA is around 40 GHz,
- BW of the first stage TIA is chosen to be 27 GHz ($n = \sqrt{2}$).
- 2 times higher $R_{F,T}$.



Trans-impedance of an inverter-based feedback TIA

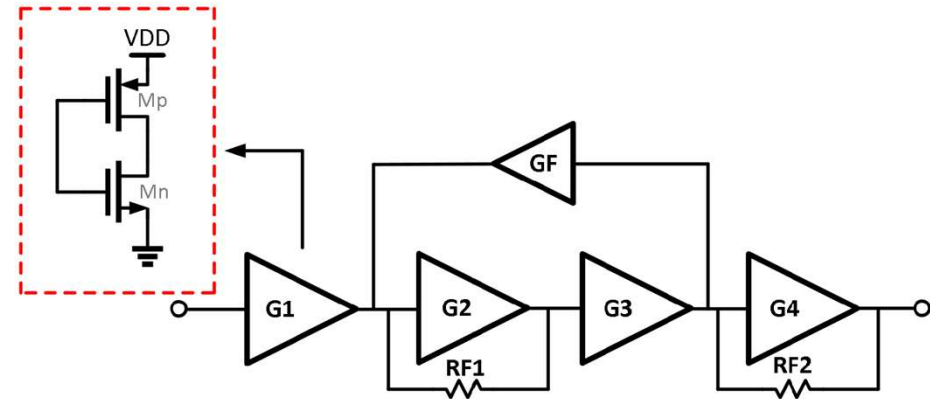
To achieve maximum bandwidth, the ratio of g_{mT}/C_{in} needs to be maximized.

$$Z_{TIA}(s) = \frac{1 - g_{mT} \cdot R_{F,T}}{s^2 C_{L,T} C_{in} R_{F,T} + s(C_{L,T} + C_{in}) + g_{mT}}$$

Design Method- TIA with AFB

Core circuit with AFB

- size of G_F transistors are small and their loading on other units is negligible.



Block diagram of the core of TIA with Active feedback.

Transfer function of each stage

$$G_1(s) = \frac{-g_{m1}(1 + s.R_{F1}.C_2)}{s^2.R_{F1}.C_1.C_2 + s.(C_1 + C_2) + g_{m2}}$$

$$G_2(s) = \frac{1 - g_{m2}.R_{F1}}{1 + s.R_{F1}.C_2}$$

$$G_3(s) = \frac{-g_{m3}(1 + s.R_{F2}.C_4)}{s^2.R_{F2}.C_3.C_4 + s.(C_3 + C_4) + g_{m4}}$$

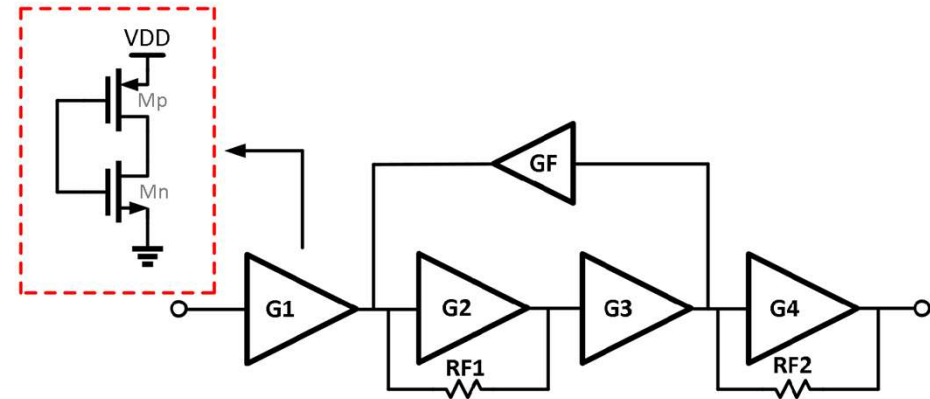
$$G_4(s) = \frac{1 - g_{m4}.R_{F2}}{1 + s.R_{F2}.C_4}$$

$$G_F(s) = \frac{-g_{mf}(1 + s.R_{F1}.C_2)}{s^2.R_{F1}.C_1.C_2 + s.(C_1 + C_2) + g_{m2}}$$

Design Method- TIA with AFB

Core circuit with AFB

$$H(s) = G_1(s) \cdot \frac{G_2(s) \cdot G_3(s)}{1 - G_2(s) \cdot G_3(s) \cdot G_F(s)} \cdot G_4(s) = \frac{N(s)}{D(s)}$$



Block diagram of the core of TIA with Active feedback.

$$D(s) = (s^2 \cdot R_{F2} \cdot C_3 \cdot C_4 + s \cdot (C_3 + C_4) + g_{m4}) \times (s^2 \cdot R_{F1} \cdot C_1 \cdot C_2 + s \cdot (C_1 + C_2) + g_{m2}) - g_{m3} \cdot g_{mf} \times (1 - g_{m2} \cdot R_{F1}) \cdot (1 + s \cdot R_{F2} \cdot C_4)$$

$$N(s) = g_{m1} \cdot g_{m3} \cdot (1 - g_{m2} \cdot R_{F1}) \cdot (1 - g_{m4} \cdot R_{F2})$$

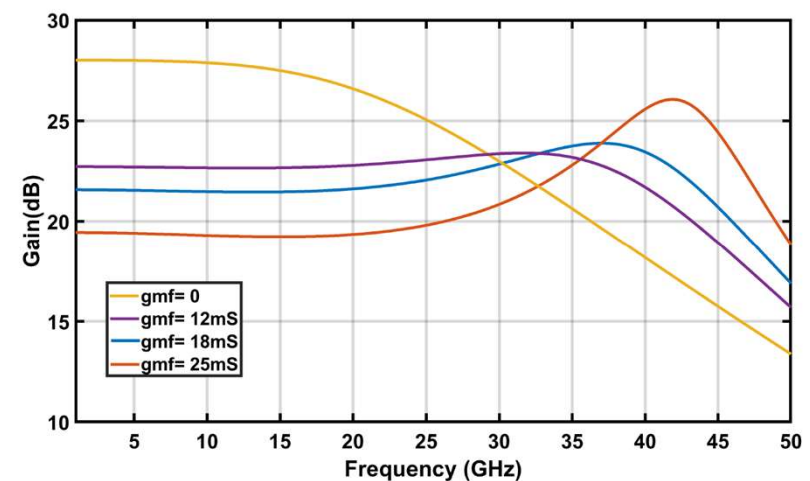
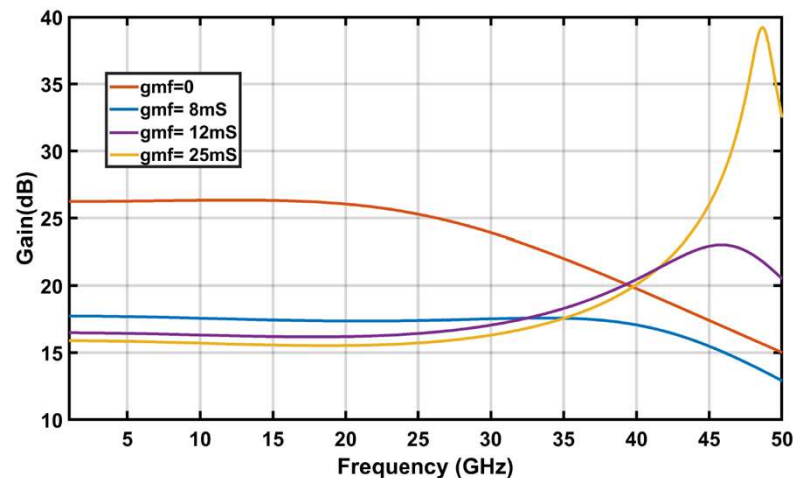
Low Frequency Gain:

$$H(0) = \frac{g_{m1} \cdot g_{m3} \cdot (1 - g_{m2} \cdot R_{F1}) \cdot (1 - g_{m4} \cdot R_{F2})}{g_{m4} \cdot g_{m2} - g_{m3} \cdot g_{mf} \cdot (1 - g_{m2} \cdot R_{F1})}$$

Design Method- TIA with AFB



- gain variations, gain peaking, and bandwidth change with different values of g_{mf} .
- Transfer function of the core circuit for 4 different values of g_{mf} for 2 different cases.
- Increasing g_{mf} reduces the gain and increases its bandwidth by making a gain peaking at high frequencies.



Design Method- TIA with AFB



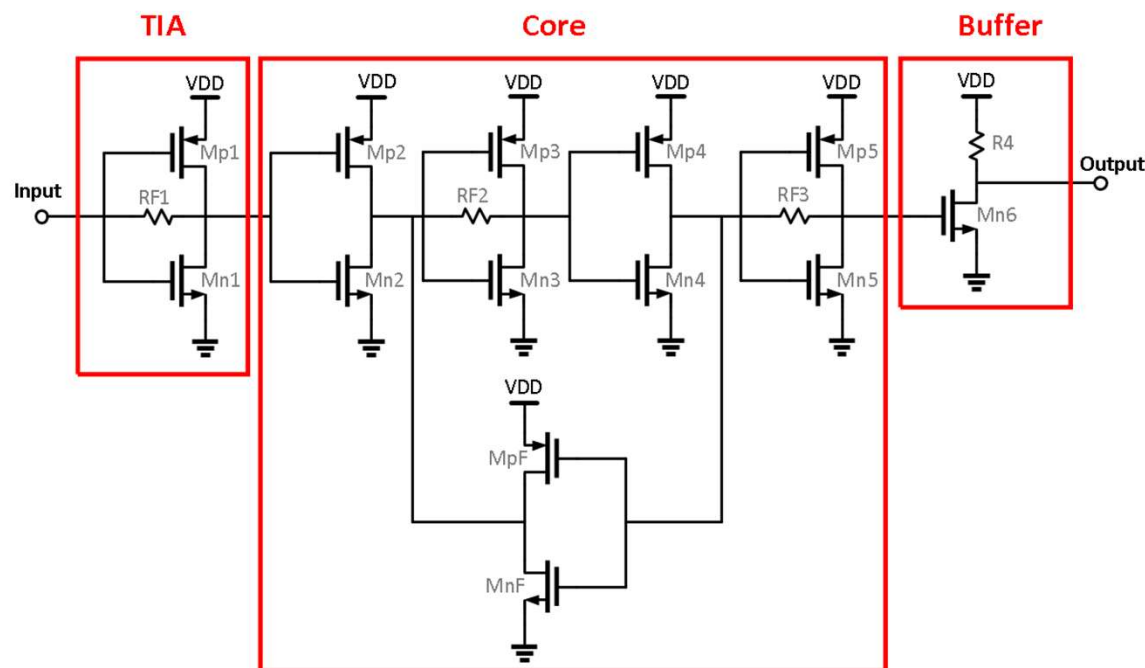
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- For $g_{mf}=0$ (no active feedback), the gain is maximum and the bandwidth is minimum.
- By increasing the value of g_{mf} , the gain decreases while at the same time the BW increases by making a peak at higher frequencies.
- As the value of g_{mf} increases, the peak value would be larger affecting the TIA's stability.
- Changing the circuit parameters shows the freedom to control low-frequency gain, high-frequency gain peaking and the bandwidth by on adjusting these values.

Design Method- TIA with AFB



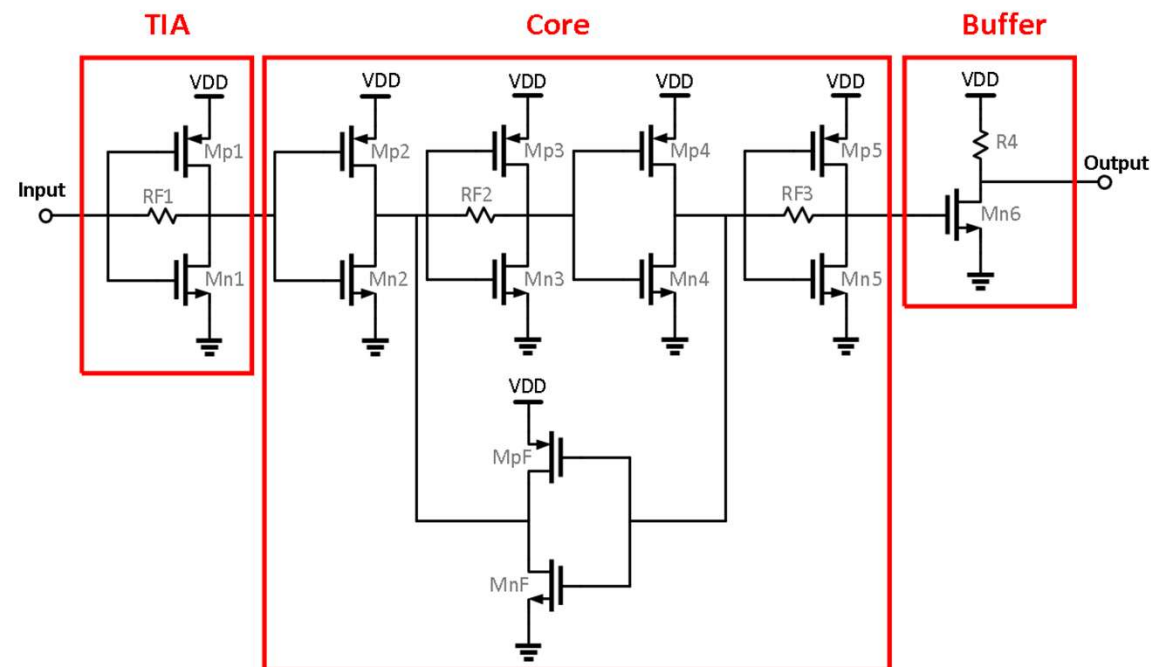
- *22 nm FDSOI SLVT*: lower threshold voltage and faster operation.
- input matching: M_{n1} and M_{p1} sizing and adjusting the value of R_{F1} .
- to minimize the noise and maximize gain: transistors are biased at the current density $J_{P,FT} = 0.25mA/um$ by adjusting their back-gate voltages, V_{bgn} and V_{bgp} .



Design Method- TIA with AFB



- The TIA implemented as a self-biased inverter through the feedback resistor which is biased at mid-supply to achieve maximum linearity.
- The size of the AFB inverters is swept for maximum bandwidth with a flat response.





Design Method- TIA with AFB

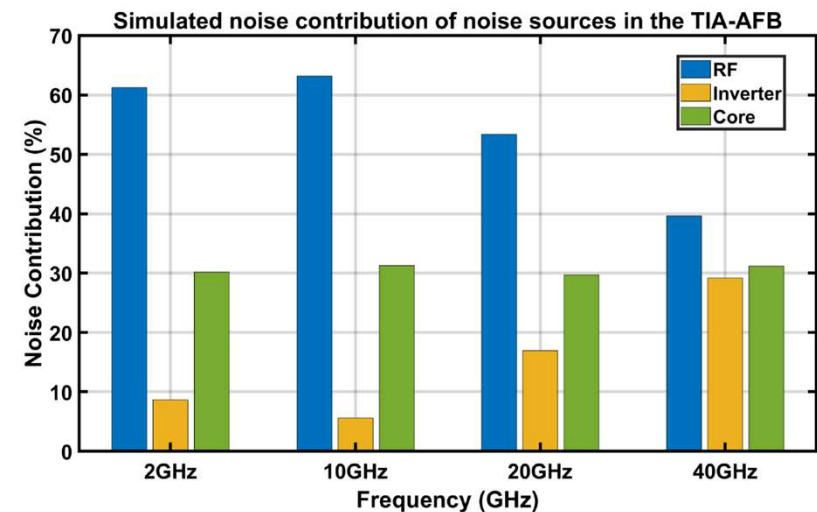
- The low-frequency input-referred current noise of the TIA-AFB is approximated as

$$\begin{aligned} \overline{I_{n,in(TIA-AFB)}^2} = & \frac{4kT}{R_{F,T}} + \frac{4kT\gamma}{g_{mT}} \cdot \frac{1}{R_{F,T}^2} + \frac{4kT\gamma}{g_{m1}} \cdot \frac{1}{A_1^2 \cdot R_{F,T}^2} + \frac{4kT\gamma}{g_{m2}} \cdot \frac{1}{A_1^2 \cdot A_2^2 \cdot R_{F,T}^2} + \\ & 4kT R_{F,1} \cdot \frac{1}{A_1^2 \cdot A_2^2 \cdot R_{F,T}^2} + \frac{4kT\gamma}{g_{m3}} * \frac{1}{A_1^2 \cdot A_2^2 \cdot A_3^2 \cdot R_{F,T}^2} + \\ & \frac{4kT\gamma}{g_{m4}} \cdot \frac{1}{Z_{TIA-AFB}^2} + 4kT R_{F,2} \cdot \frac{1}{Z_{TIA-AFB}^2} + \frac{4kT\gamma}{g_{mF}} \cdot \frac{1}{A_1^2 \cdot R_{F,T}^2} \end{aligned}$$

Design Method- TIA with AFB

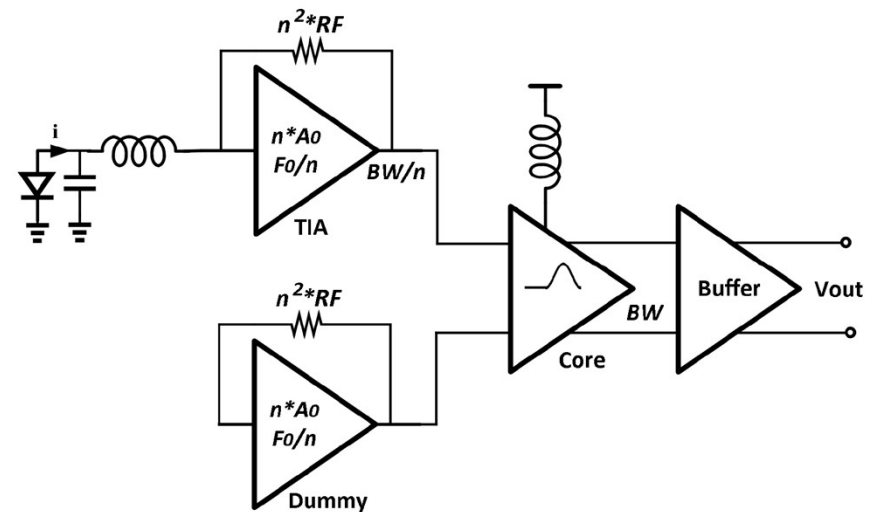


- The primary source of noise at low frequencies is $R_{F,T}$ of the narrow-band inverter-based TIA.
- The inverter of narrow-band TIA, characterized by a g_{mT} , introduces more noise at the input compared to other inverter stages G_i , as other noise sources are divided by the gain factor of previous stages.



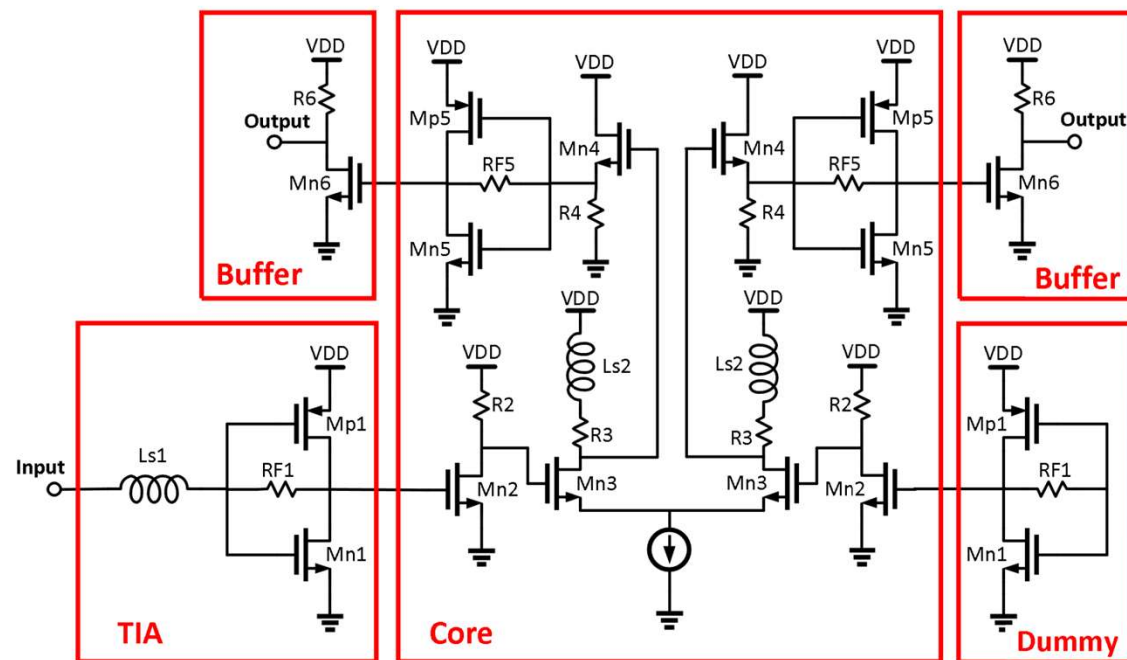
Design Method-TIA with Inductive Peaking

- Narrow-band inverter-based TIA: similar to the first stage of TIA-AFB with narrow bandwidth.
- core with inductive peaking: shunt peaking.
- Series inductor at the first stage.



Design Method-TIA with Inductive Peaking

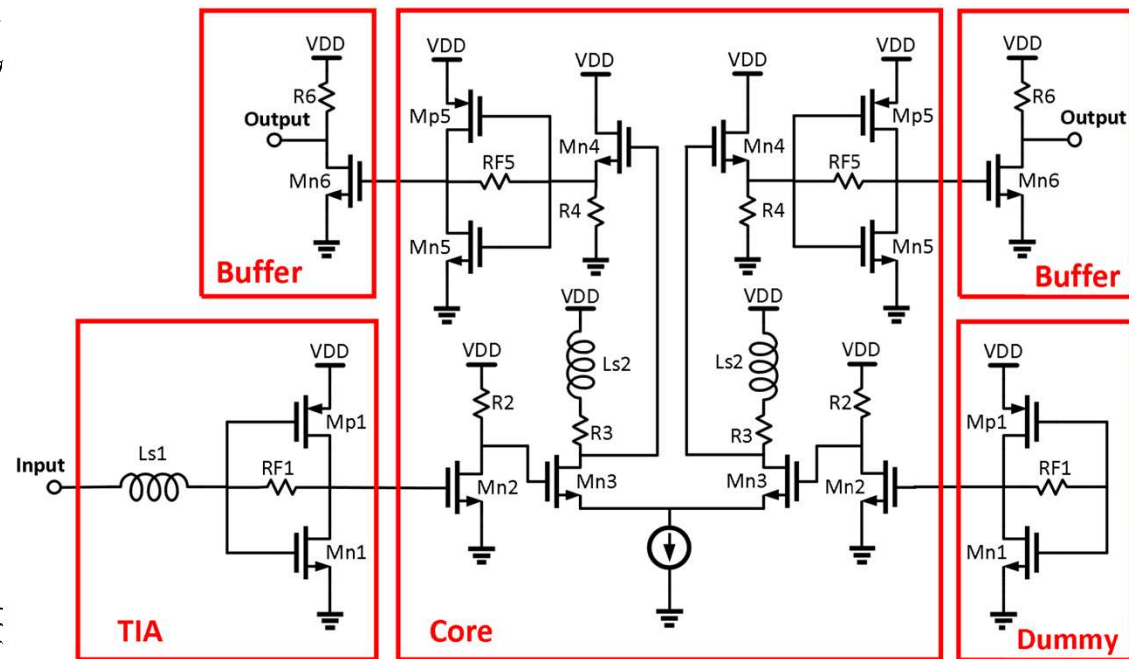
- The BW limitation: input stage parasitic and photo diode capacitance is compensated using series inductive peaking $Ls1$ and providing a peaking at higher frequencies.
- Shunt peaking increases the BW: introduces a zero in transfer function.
- The bandwidth enhancement ratio (BWER) of 1.85 for a cascaded (CS) stage is feasible.



Schematic of TIA with Inductive peaking

Design Method-TIA with Inductive Peaking

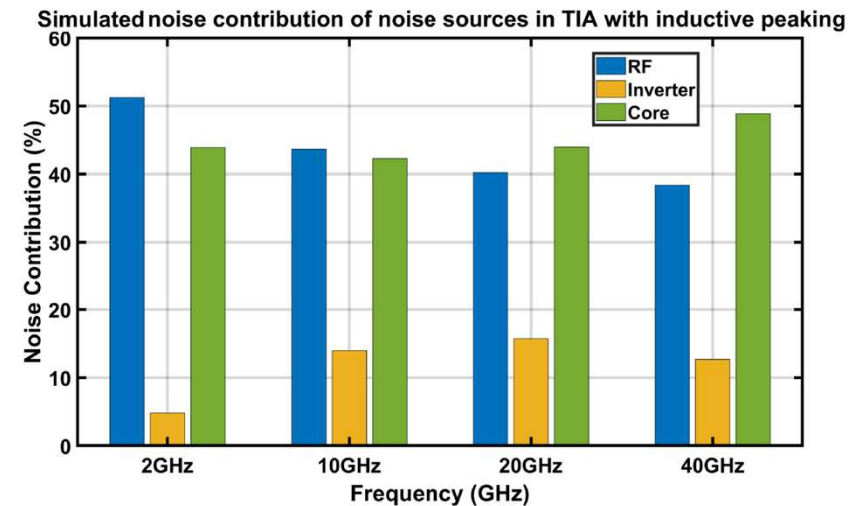
- The first stage: 24 GHz bandwidth, $n = \sqrt{2}$: the bandwidth is scaled down by a factor of $\sqrt{2}$
- The core part restores the bandwidth by means of shunt peaking to around 33 GHz.
- The series inductor and the PD capacitor model which is around 70 fF at the input makes a peak around 40 GHz.



Schematic of TIA with Inductive peaking

Design Method-TIA with Inductive Peaking

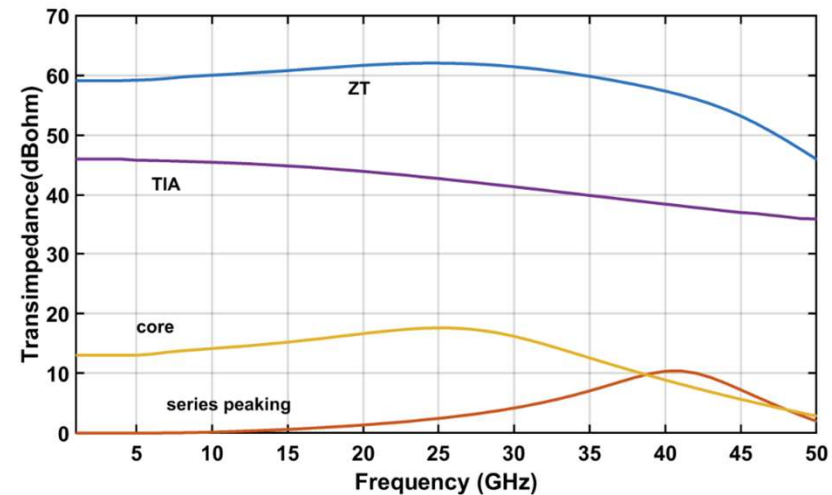
- The bandwidth decreases by a factor of $n = \sqrt{2}$, the white noise due to $R_{F,T}$ decreases by a factor of 2.
- Additionally, the white noise of the amplifiers decreases by $n^4 = 4$.
- This reduction proportional to n^2 and n^4 can provide sufficient noise suppression.



Simulated noise contribution of noise sources in TIA with inductive peaking.

Design Method-TIA with Inductive Peaking

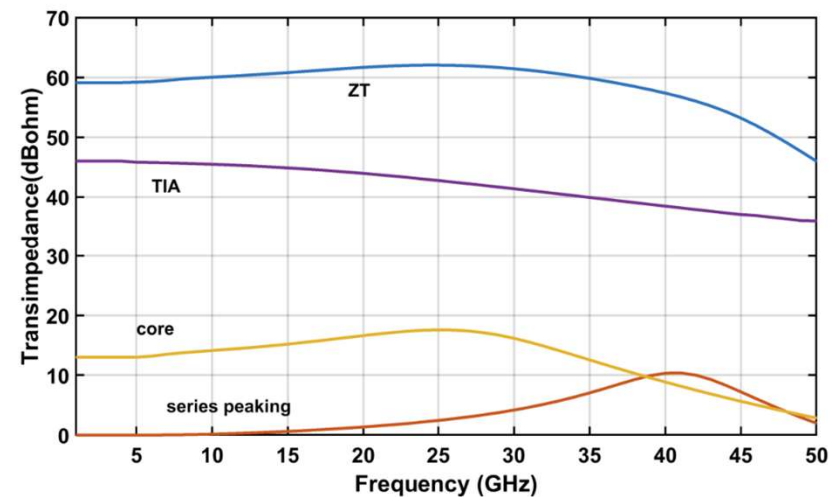
- The first stage TIA alone has around 24 GHz bandwidth and the core part restores the bandwidth by means of shunt peaking to around 33 GHz. The series inductor and the PD capacitor which is 50 fF at the input makes a peak around 40 GHz.
- The TIA provide a 66 dBΩ differential transimpedance over 40 GHz bandwidth.
- In this design $n = 1.4$, which means the bandwidth is scaled down by a factor of 1.4 to around 24 GHz for the first stage and RF,T is scaled up by a factor of $n^2 = 2$.



Design Method



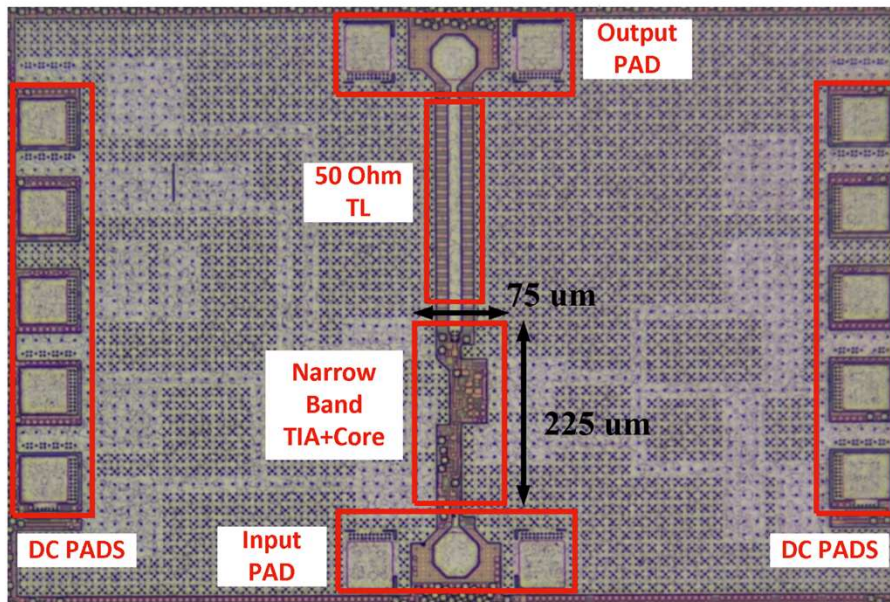
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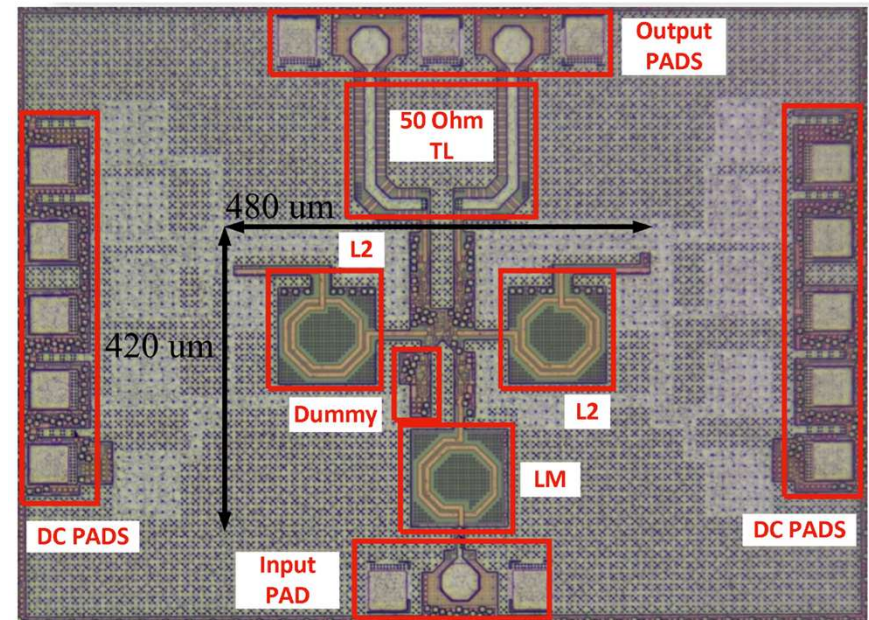
LAYOUT IMPLEMENTATION AND FABRICATION



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TIA with AFB fabricated on CMOS 22nm CMOS FDSOI

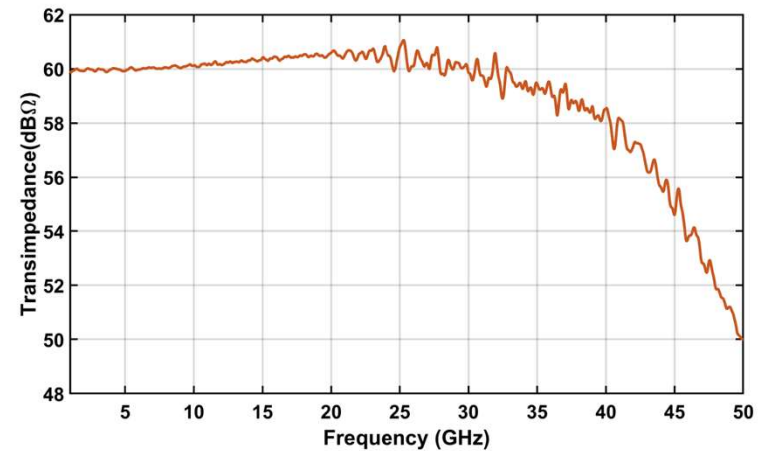
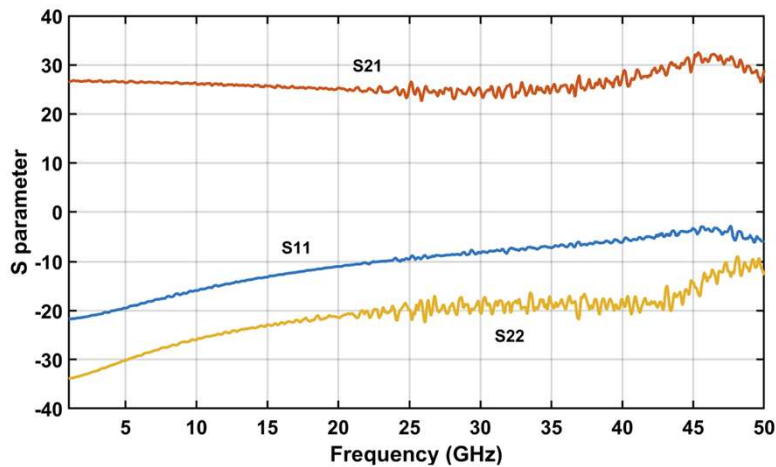


TIA with Inductive peaking fabricated on CMOS 22nm CMOS FDSOI



Measurements- TIA with AFB

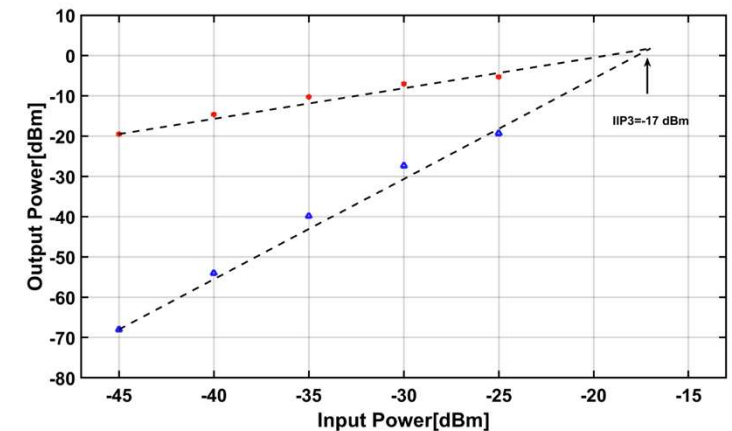
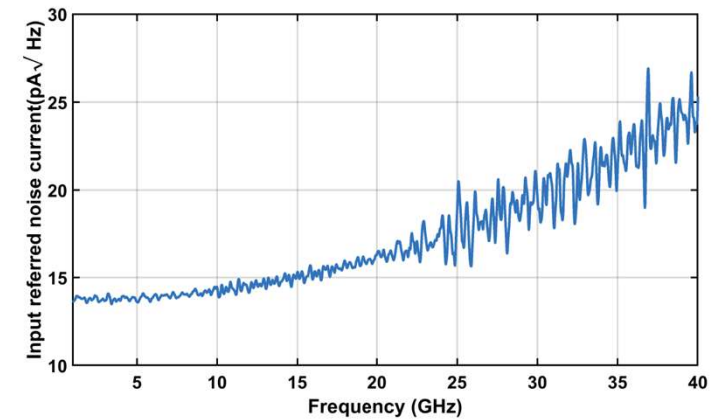
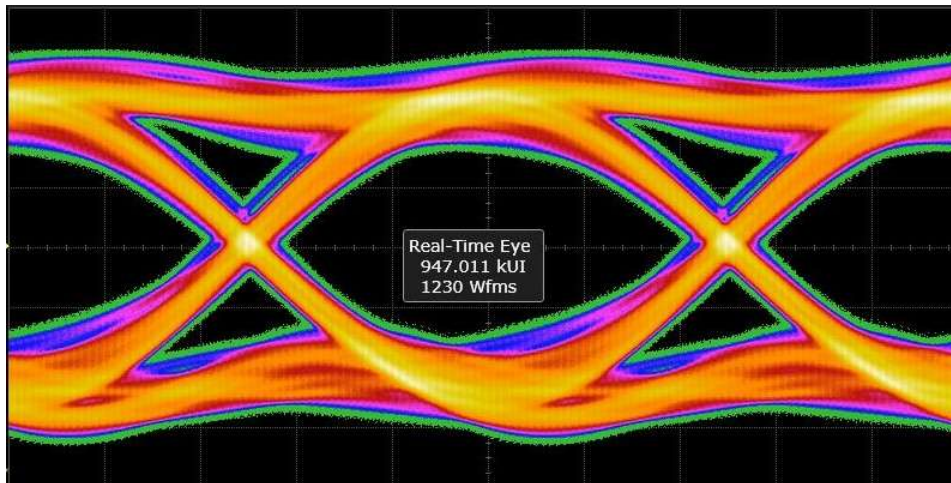
- The S11 and S22 for AFB-TIA are measured to be lower than -6dB and -17dB . S21 is measured to be about 27dB .
- The PD is modeled by 50fF capacitor.
- The TIA provides a transimpedance gain of $60\text{dB}\Omega$ over 40GHz bandwidth.





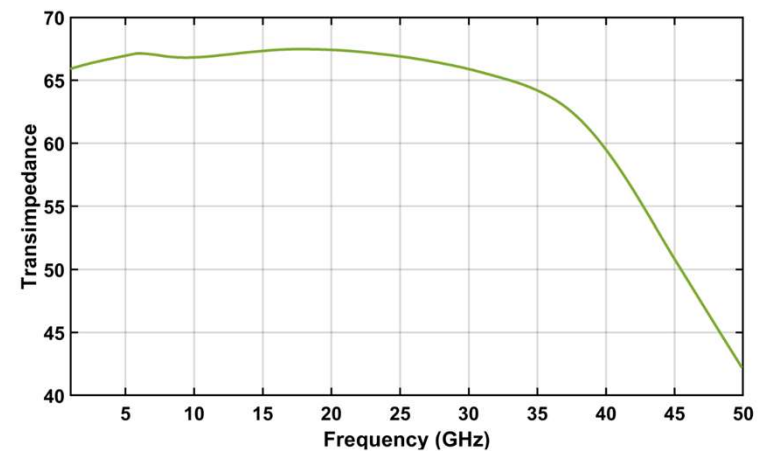
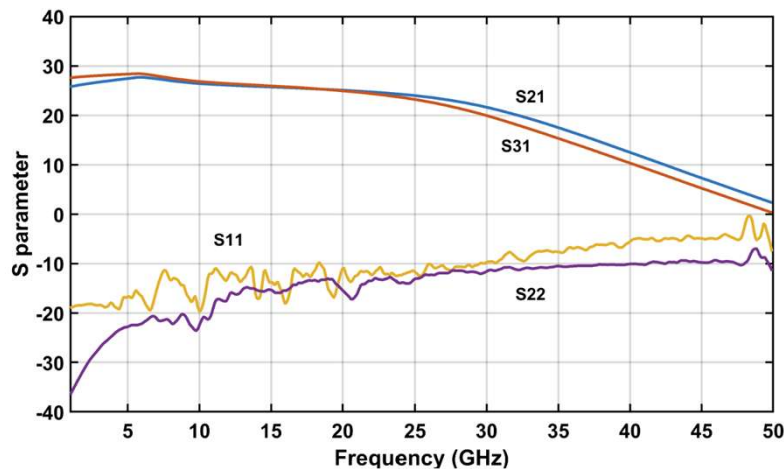
Measurements- TIA with AFB

- input referred current noise density is $16\text{pA}/\sqrt{\text{Hz}}$ at mid-band.
- IIP3 at 20GHz is around -17 dBm.
- Eye diagram for 40 Gb/s.



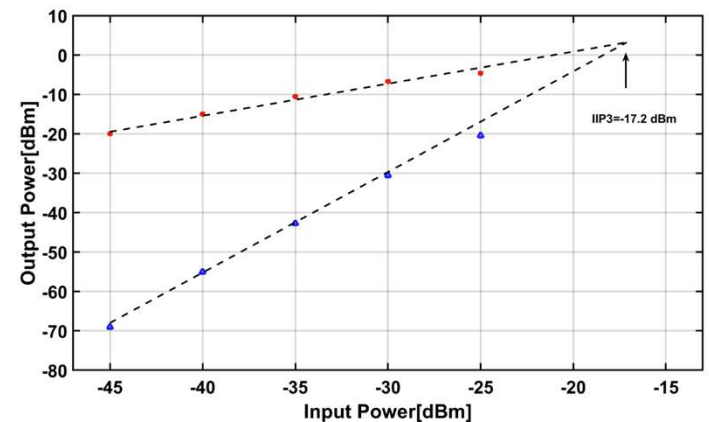
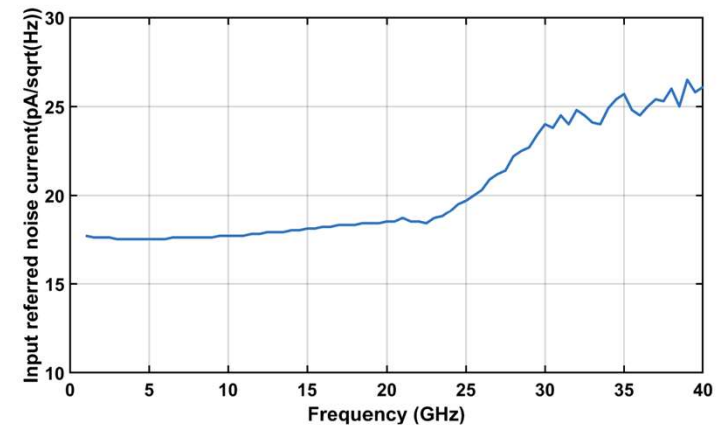
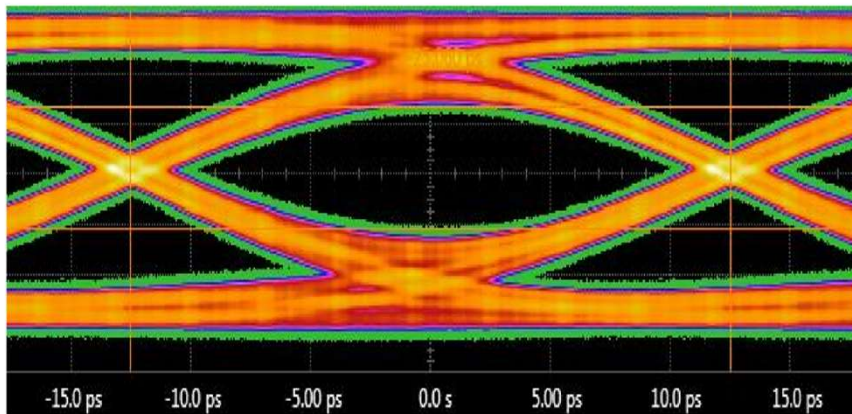
Measurements- TIA with Inductive peaking

- The S11 and S22 for AFB-TIA are measured to be lower than -7dB and -10dB . S21 is measured to be about 26dB single ended.
- The PD is modeled by 70fF capacitor.
- The TIA provides a transimpedance gain of $66\text{dB}\Omega$ over 37GHz bandwidth.



Measurements- TIA with Inductive peaking

- input referred current noise density is $18\text{pA}/\sqrt{\text{Hz}}$ at mid-band.
- IIP3 at 20GHz is around -17.2 dBm.
- Eye diagram for 38 Gb/s.





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